

TCS IPA IT-IS Stream Networking

Top 50 MCQ's

Q1. A user can access websites using IP addresses, but domain names like `www.example.com` do not work. Which service is most likely having an issue? (Easy)

- A) DHCP
- B) DNS ☒
- C) FTP
- D) SMTP

Memory Booster:

DNS converts domain names into IP addresses. If DNS fails, websites can still be reached by typing their IP addresses. DHCP only assigns IP addresses—it doesn't resolve website names.

Q2. An IT engineer wants to connect multiple computers in the same office so they can communicate efficiently without unnecessary network traffic. Which device should be used? (Easy)

- A) Hub
- B) Switch ☒
- C) Modem
- D) Repeater

Memory Booster:

A switch sends data only to the intended device, improving speed and reducing collisions. A hub sends data to every connected device, creating unnecessary traffic.

Q3. Which command is commonly used in Windows to verify whether a remote server is reachable over the network? (Easy)

- A) format
- B) ping ☒
- C) copy
- D) shutdown

Memory Booster:

ping checks connectivity by sending ICMP Echo Requests. If replies are received, the destination is reachable. It is usually the first command used during network troubleshooting.

Q4. A laptop is configured to obtain its IP address automatically, but it receives an address starting with 169.254.x.x. What is the most likely cause? (Medium)

- A) DNS server failure
- B) DHCP server is unavailable ☒
- C) Default gateway is incorrect
- D) Firewall is blocking traffic

Memory Booster:

A 169.254.x.x address (APIPA) is assigned when the device cannot contact a DHCP server. This allows limited local communication but not normal network access.

Q5. A Linux administrator wants to securely log in to a remote server for maintenance. Which protocol should be used? (Medium)

- A) Telnet
- B) SSH ☒
- C) FTP
- D) HTTP

Memory Booster:

SSH encrypts all communication, making remote administration secure. Telnet sends usernames and passwords in plain text, making it unsafe for modern networks.

Q6. A user reports that they can access computers within the office but cannot browse the internet. Which network setting should the administrator check first? (Medium)

- A) Computer hostname
- B) Default gateway 
- C) MAC address
- D) Monitor resolution

Memory Booster:

The default gateway is the router that forwards traffic outside the local network. If it's missing or incorrect, local communication works, but internet access usually fails.

Q7. According to ITIL, a user reports that they cannot connect to the company Wi-Fi. What should the Service Desk do first? (Medium)

- A) Replace the wireless access point immediately
- B) Log the incident and begin initial diagnosis 
- C) Escalate directly to senior management
- D) Reinstall the operating system

Memory Booster:

ITIL recommends logging every incident before troubleshooting. Proper documentation helps track issues, identify trends, and improve support quality over time.

Q8. A company wants to divide one large network into several smaller networks to reduce broadcast traffic and improve performance. Which technology should be used? (Difficult)

- A) VLAN 
- B) NAT
- C) Proxy Server
- D) Load Balancer

Memory Booster:

A VLAN logically separates devices into different broadcast domains even if they're connected to the same switch. This improves security and reduces unnecessary broadcasts.

Q9. An administrator configures a subnet mask of 255.255.255.0. Approximately how many usable host IP addresses are available in that subnet? (Difficult)

- A) 128
- B) 254 ☒
- C) 256
- D) 512

Memory Booster:

A /24 network has 256 total addresses. Two are reserved—one for the network ID and one for the broadcast address—leaving **254 usable host addresses**.

Q10. During troubleshooting, a network engineer notices that devices can communicate using IP addresses but fail when trying to access shared resources using computer names. Which service should be investigated first? (Difficult)

- A) DNS ☒
- B) DHCP
- C) NTP
- D) SNMP

Memory Booster:

When IP-based communication works but name-based communication fails, DNS is usually the problem because it translates names into IP addresses. DHCP assigns IPs but doesn't resolve names.

Q11. A user says that all devices in the office can access the internet except their own laptop. What should the IT engineer check first? (Easy)

- A) ISP status
- B) Laptop's network configuration ☒
- C) Company firewall rules
- D) DNS records on the internet

Memory Booster:

If only one device is affected, the problem is usually on that device. Check IP settings, Wi-Fi connection, or network adapter before investigating the entire network.

Q12. Which protocol is mainly responsible for automatically assigning IP addresses to devices on a network? (Easy)

- A) DNS
- B) DHCP 
- C) HTTPS
- D) ICMP

Memory Booster:

DHCP automatically provides IP addresses, subnet masks, gateways, and DNS server details. Without DHCP, users often need to configure network settings manually.

Q13. Which device connects two different networks and decides the best path for data packets? (Easy)

- A) Hub
- B) Switch
- C) Router 
- D) Bridge

Memory Booster:

A router forwards packets between different networks, such as your office LAN and the internet. A switch works only within the same local network.

Q14. A Windows user successfully pings 127.0.0.1, but cannot ping another computer on the same network. What does this indicate? (Medium)

- A) The local TCP/IP stack is working 
- B) The network cable is disconnected
- C) The DHCP server has failed
- D) The switch is faulty

Memory Booster:

The address **127.0.0.1** is the loopback address. A successful ping confirms that the computer's TCP/IP software is functioning, but it doesn't verify external network connectivity.

Q15. An employee reports that file transfers over the network are much slower than usual. Which command would help identify the path taken by packets to the destination? (Medium)

- A) ipconfig
- B) tracert 
- C) hostname
- D) net user

Memory Booster:

tracert (Windows) shows every router (hop) between your computer and the destination. It helps identify where delays or failures occur in the network.

Q16. A Linux administrator wants to view the IP address assigned to a network interface using a modern command. Which command should be used? (Medium)

- A) ip addr 
- B) mkdir
- C) passwd
- D) chmod

Memory Booster:

ip addr displays IP addresses and interface details on Linux. It is the modern replacement for the older ifconfig command on many Linux distributions.

Q17. While troubleshooting, an administrator notices that two computers have been assigned the same IP address. What is the most likely result? (Medium)

- A) Faster network speed
- B) IP address conflict causing connectivity issues 
- C) Automatic creation of a backup IP
- D) Improved DNS resolution

Memory Booster:

Every device on the same network must have a unique IP address. Duplicate IPs cause communication problems because the network cannot determine which device should receive the data.

Q18. A company wants internal IP addresses to remain hidden from the public internet while allowing employees to browse websites. Which technology performs this function? (Difficult)

- A) NAT ☒
- B) VLAN
- C) ICMP
- D) FTP

Memory Booster:

NAT (Network Address Translation) converts private IP addresses into a public IP before sending traffic to the internet. This conserves public IPs and improves security by hiding internal addresses.

Q19. An engineer captures packets and notices that every device connected to a particular network segment receives all transmitted frames, causing unnecessary traffic. Which device is most likely being used? (Difficult)

- A) Router
- B) Managed Switch
- C) Hub ☒
- D) Firewall

Memory Booster:

A hub broadcasts incoming data to all connected ports, regardless of the destination. A switch is smarter—it forwards frames only to the correct device, reducing unnecessary traffic.

Q20. A user can successfully ping the default gateway but cannot access any external websites. Other users on the same network have internet access. Which troubleshooting step should be performed next? (Difficult)

- A) Replace the network switch immediately
- B) Check the user's DNS server configuration ☒
- C) Restart the ISP's router
- D) Format the user's computer

Memory Booster:

If the gateway is reachable, the local network is working. Since only one user is affected, an incorrect DNS configuration is a likely cause. Without DNS, websites cannot be reached by name even though the internet connection exists.

Q21. An employee connects a new laptop to the office Wi-Fi but cannot communicate with any other devices. Which setting should be checked first? (Easy)

- A) Screen brightness
- B) IP address configuration 
- C) Printer driver
- D) Browser history

Memory Booster:

Without a valid IP address, the laptop cannot communicate on the network. Always verify whether the device has received the correct IP settings before checking anything else.

Q22. Which protocol is primarily used to transfer web pages securely over the internet? (Easy)

- A) HTTP
- B) HTTPS 
- C) FTP
- D) Telnet

Memory Booster:

HTTPS encrypts communication between your browser and the web server using SSL/TLS. This protects sensitive information like passwords and banking details.

Q23. Which command in Windows displays the current IP address, subnet mask, and default gateway? (Easy)

- A) ping
- B) ipconfig 
- C) nslookup
- D) netstat

Memory Booster:

ipconfig quickly shows your network configuration. It's one of the first commands used by support engineers during network troubleshooting.

Q24. A network administrator changes the DNS server on a user's PC. The user still cannot access websites by name because old DNS records are being used. Which command should be executed? (Medium)

- A) ipconfig /flushdns 
- B) format C:
- C) shutdown /r
- D) tracert

Memory Booster:

ipconfig /flushdns clears the local DNS cache so the computer requests fresh records from the DNS server. This is useful after DNS changes.

Q25. A user reports that email is working, but shared folders on another computer cannot be opened using its hostname. However, accessing the shared folder using the IP address works. What is the most likely issue? (Medium)

- A) DNS name resolution problem 
- B) Faulty Ethernet cable
- C) DHCP lease renewal failure
- D) Antivirus software corruption

Memory Booster:

If access using the IP address works but the hostname fails, the network connection is fine. The problem is usually with name resolution (DNS).

Q26. Which Linux command is commonly used to test network connectivity to another host? (Medium)

- A) ls
- B) ping ☒
- C) pwd
- D) grep

Memory Booster:

ping works in both Linux and Windows. It helps verify whether another host is reachable over the network before performing advanced troubleshooting.

Q27. During troubleshooting, an engineer notices that the link light on a network switch port is OFF after connecting a desktop. What is the most likely cause? (Medium)

- A) Physical connection problem ☒
- B) Incorrect DNS entry
- C) Browser cache issue
- D) Wrong system time

Memory Booster:

If the link LED is off, the switch does not detect a physical connection. Check the Ethernet cable, NIC, or switch port before investigating software issues.

Q28. A company wants employees in different departments to be isolated from each other while using the same physical switch infrastructure. Which solution is most appropriate? (Difficult)

- A) VLAN ☒
- B) DHCP Relay
- C) FTP Server
- D) Proxy Cache

Memory Booster:

VLANs create separate logical networks on the same switch. For example, HR and Finance can use the same switch while remaining isolated for security.

Q29. An administrator runs the netstat command and notices many connections in the LISTENING state. What does this indicate? (Difficult)

- A) The computer is waiting for incoming network connections 
- B) The network cable is disconnected
- C) DNS is not working
- D) The system has no IP address

Memory Booster:

A **LISTENING** port means a service is ready to accept incoming connections, such as a web server waiting for client requests. It does not necessarily indicate a problem.

Q30. A company has multiple branch offices connected through routers. An engineer wants to verify every network device (hop) that packets pass through before reaching a remote server. Which utility should be used? (Difficult)

- A) nslookup
- B) tracer 
- C) hostname
- D) tasklist

Memory Booster:

tracer displays each router (hop) between the source and destination. If packets stop at a particular hop, that device or network segment is likely causing the issue.

Q31. A desktop shows "Network cable unplugged" even though Wi-Fi is turned off. What should the technician check first? (Easy)

- A) Ethernet cable connection 
- B) DNS server settings
- C) Browser cache
- D) User password

Memory Booster:

This message usually indicates a physical connectivity issue. Check whether the Ethernet cable is properly connected or if the cable/NIC is damaged before checking software settings.

Q32. Which device converts digital signals from a computer into signals suitable for transmission over an ISP connection? (Easy)

- A) Switch
- B) Router
- C) Modem ☒
- D) Hub

Memory Booster:

A modem connects your home or office network to the ISP. The router then distributes the internet connection to multiple devices.

Q33. Which protocol is commonly used to automatically synchronize the system time of computers over a network? (Easy)

- A) FTP
- B) NTP ☒
- C) SMTP
- D) SNMP

Memory Booster:

NTP (Network Time Protocol) keeps system clocks accurate. Correct time is essential for authentication, logging, and security certificates.

Q34. A Windows user cannot browse websites. The administrator suspects a DNS issue. Which command can be used to directly query a DNS server? (Medium)

- A) ipconfig
- B) nslookup ☒
- C) netstat
- D) hostname

Memory Booster:

nslookup checks whether a DNS server can resolve a domain name into an IP address. If it fails, the issue is likely with DNS rather than internet connectivity.

Q35. An office has a printer with the IP address 192.168.1.50. A user's PC has the IP 192.168.2.20 with the subnet mask 255.255.255.0. They cannot communicate. What is the most likely reason? (Medium)

- A) They are in different subnets ☒
- B) DNS is unavailable
- C) Printer is using HTTPS
- D) The PC has too much RAM

Memory Booster:

With a **255.255.255.0** mask, devices must have the same first three octets to communicate directly. Here, **192.168.1.x** and **192.168.2.x** belong to different subnets.

Q36. An IT engineer wants to view all active network connections and listening ports on a Windows computer. Which command should be used? (Medium)

- A) netstat ☒
- B) ipconfig
- C) whoami
- D) cls

Memory Booster:

netstat displays active TCP/UDP connections and listening ports. It is useful for identifying unexpected network activity or troubleshooting services.

Q37. A user can successfully ping the DNS server but cannot access any website. nslookup also fails to resolve domain names. What is the most likely issue? (Medium)

- A) DNS service is not responding ☒
- B) Network cable is unplugged
- C) Switch has failed completely
- D) Monitor driver is outdated

Memory Booster:

If the DNS server is reachable but cannot resolve names, the DNS service itself may be down or misconfigured. Network connectivity exists, but name resolution does not.

Q38. A network administrator wants to identify the MAC address associated with an IP address on the local network. Which command is most suitable? (Difficult)

- A) arp -a ☒
- B) ipconfig
- C) ping
- D) tracert

Memory Booster:

arp -a displays the ARP table, showing mappings between IP addresses and MAC addresses. ARP works only within the local network segment.

Q39. A company deploys a firewall between its internal network and the internet. What is the firewall's primary purpose? (Difficult)

- A) Increase monitor resolution
- B) Filter and control network traffic ☒
- C) Assign IP addresses automatically
- D) Resolve domain names

Memory Booster:

A firewall allows or blocks traffic based on security rules. It acts like a security guard at the entrance of a building, deciding who can enter or leave.

Q40. A server hosts both a website and an SSH service. Which combination of default ports is correct? (Difficult)

- A) HTTP – 22, SSH – 80
- B) HTTP – 21, SSH – 23
- C) HTTP – 80, SSH – 22 ☒
- D) HTTP – 25, SSH – 53

Memory Booster:

Remember these common ports:

- HTTP → 80
- HTTPS → 443
- SSH → 22
- FTP → 21

- **DNS → 53**

These are among the most frequently asked port numbers in TCS IPA.

Q41. A user reports that they can access websites but cannot send emails using an email client like Outlook. Web browsing works normally. Which service should the administrator check first? (Easy)

- A) SMTP service ☒
- B) DHCP server
- C) NTP server
- D) ARP table

Memory Booster:

SMTP is responsible for sending emails. If browsing works but emails cannot be sent, the issue is likely with the SMTP server or its configuration.

Q42. Which protocol is mainly used to transfer files between a client and a server? (Easy)

- A) FTP ☒
- B) DNS
- C) ICMP
- D) SNMP

Memory Booster:

FTP (File Transfer Protocol) is specifically designed for file transfers. HTTP delivers web pages, while DNS resolves names into IP addresses.

Q43. Which command in Linux is commonly used to display the routing table? (Easy)

- A) route ☒
- B) ls
- C) cat
- D) pwd

Memory Booster:

The route command (or ip route in modern Linux) shows how packets are forwarded to different networks. It's useful when troubleshooting gateway-related issues.

Q44. A network engineer notices that users in one department cannot communicate with another department after VLANs were configured. Which device is required to enable communication between different VLANs? (Medium)

- A) Hub
- B) Layer 3 Device (Router/Layer 3 Switch) ☒
- C) Repeater
- D) Access Point

Memory Booster:

Devices in different VLANs are in different networks. A router or Layer 3 switch performs **inter-VLAN routing** so they can communicate securely.

Q45. A Windows PC has the correct IP address and subnet mask but cannot access devices outside its network. Which setting is most likely missing? (Medium)

- A) Default Gateway ☒
- B) MAC Address
- C) Hostname
- D) Workgroup Name

Memory Booster:

The default gateway is the "exit door" to other networks. Without it, the PC can communicate only within its own subnet.

Q46. A user reports intermittent network connectivity. Which command continuously sends packets until manually stopped, helping identify packet loss over time? (Medium)

- A) ping -t ☒
- B) ipconfig /release
- C) netstat -a
- D) hostname

Memory Booster:

In Windows, ping -t keeps sending ping requests until stopped with **Ctrl + C**. It's useful for detecting unstable network connections.

Q47. While checking a switch, an engineer finds that one port is showing excessive collisions. Which device is most likely connected to that port? (Medium)

- A) Hub ☒
- B) Managed Switch
- C) Router
- D) Firewall

Memory Booster:

Hubs operate in half-duplex mode and share bandwidth, leading to collisions. Modern switches use full-duplex communication, which eliminates collisions.

Q48. A company wants to prevent unauthorized devices from obtaining IP addresses by connecting to the office network. Which network service should be secured? (Difficult)

- A) DHCP ☒
- B) DNS
- C) HTTP
- D) ICMP

Memory Booster:

DHCP assigns IP addresses automatically. Securing DHCP (using features like DHCP Snooping) helps prevent rogue DHCP servers and unauthorized devices from disrupting the network.

Q49. During troubleshooting, an administrator finds that a server responds to ping but no users can open the hosted website. Which is the most likely reason? (Difficult)

- A) Web service is stopped ☒
- B) Ethernet cable is disconnected
- C) DHCP server is unavailable
- D) Switch is powered off

Memory Booster:

A successful ping confirms the server is reachable on the network. If the website is unavailable, the web service (such as IIS or Apache) may not be running.

Q50. A support engineer receives an incident where multiple users suddenly lose network connectivity after a switch reboot. According to ITIL, what should the engineer do after restoring the service? (Difficult)

- A) Close the incident without documentation
- B) Record the resolution and update the incident ticket 
- C) Immediately replace all switches
- D) Delete the system logs

Memory Booster:

ITIL emphasizes proper documentation after resolving an incident. Recording the root cause and resolution helps future troubleshooting and builds the organization's knowledge base.